ALLEGRA WONG

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EDUCATION

The Cooper Union for the Advancement of Science and Art | New York, NY — Expected May 2028
Bachelor of Mechanical Engineering | GPA: 3.49
Relevant Coursework: Mechanics of Materials, Dynamics, Materials Science, Statics, Intro to Finite Element Analysis, Design and Prototyping, Dynamic Systems Modeling
Awards: Half-Tuition Scholarship, Samuel and Sally Gilman Scholarship Fund, Cooper Hawks Grant

WORK EXPERIENCE

Lab Tech Assistant — May 2026 – Present
Cooper Union | New York, NY
- Operate and supervise laser cutters and 3D printers, assisting engineering students with rapid prototyping, material selection, and fabrication.
- Design organizational and standardization needs for mechanical engineering department.

Product Design and Energy Modeling Intern — Mar. – Aug. 2026
The Loisaida Center | New York, NY
- Designed air quality monitors, including developing 3D models, sourcing electronics, fabricating laser-cut parts, and assembling prototypes for batch production.
- Improved assembly workflow within a four-person team to reduce production time by approximately 60%, supporting deployment of 20 units and development of plans to scale production to 100 units.
- Developed building energy and CFD models using Rhino Grasshopper and Ansys Workbench to investigate ventilation and energy retrofit strategies for an existing facility.
- Analyzed integration of energy recovery ventilator in Loisaida to study indoor air quality and energy use.

Waitress/Hostess — Jul. – August 2025
The Bao | New York, NY
- Managed front-of-house operations, including takeout orders and kitchen deliveries, in a fast-paced environment.
- Exercised strong communication and collaboration with staff and patrons to maintain smooth and efficient service.

ACADEMIC PROJECTS

Brake Design Lead — Aug. 2026 – Present
Cooper Union Toboggan (GNCTR) Team | New York, NY
- Design snowplow break, performing hand calculations under full load conditions to validate FEA simulations.
- Iteratively 3D model functional components, incorporating GD&T standards for machining and welding.

Research Assistant — Sep. 2025 – Jul. 2026
Bio-Inspired Design Lab | New York, NY
- Co-designed and fabricated a low-cost, open-source aerial micro robot within a three-person team, developing CAD models, rapid prototyping, and assembly instructions.
- Modeled mechanical transmission systems in Fusion 360 to support motor feedback and developed assembly instructions to improve reproducibility and support future iterations.

LEADERSHIP AND INVOLVEMENT

Society of Women's Engineers (SWE) | *Event Coordinator* | *Cooper Union* — May 2025 – Present
Chinese Student Association (CSA) | *Board Member* | *Cooper Union* — May 2026 – Present
American Society of Mechanical Engineers (ASME) | *Social Media* | *Cooper Union* — Feb. 2025 – Present

TECHNICAL SKILLS

CAD and Simulation: Onshape, Fusion360, SolidWorks, GD&T, AutoCAD, ANSYS ADPL
Fabrication: 3D printing, Laser Cutting, Machining Tools
Software: MATLAB, LaTeX, Microsoft Office, Googe Workspace, Canva